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TOOL FOR PREDICTING PLANS AND IMPROVING SOLID WASTE MANAGEMENT BY APPLYING TIME SERIES AND AWS

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***Abstract*—Currently, the constant population growth in Peru leads to the generation of large amounts of solid waste, most of these originate from domestic sources and are collected by municipalities. The inefficiency in solid waste collection plans generates rework and therefore operational overhead. Likewise, it pollutes the environment due to waste accumulated in the streets and dissatisfaction from the neighbors. Therefore, an IoT tool is proposed, which will have a solar panel, sensor, chip inside the containers that served to train the algorithm and predict solid waste collection plans for decision-making of a municipality using the time series model, arima and Random Forest. This tool has 3 stages: analysis, design, and validation. Stage 1 evaluates machine learning models and IoT architectures. Stage 2 describes the design and software architecture of the tool. And, in stage 3, the deployment of IoT devices in a municipality of Lima was simulated and its operation was validated with the solid waste collection area. The result of the implementation of the tool in a sector of Villa El Salvador managed to increase the efficiency of solid waste collection plans from 81% to 96% effectiveness in said sector. In addition, this reduced operational costs by approximately 8% and increased the satisfaction of neighbors by not having solid waste accumulated in the streets.**

***Keywords*—sensors, predictive algorithms, path , Machine learning, solid waste.**

I. Introduction

Smart cities are a complex system capable of using information technologies (IT) to create infrastructure and services offered to citizens. These cities aim to create an intelligent environment that helps improve the quality of life of their citizens, from public transportation to energy savings and efficiency. The IoT is also a system of devices connected to IT that aid in communication with the internet [1].

With the growth of the world population and the constant demand for food and other materials, there is an increase in the amount of waste generated daily by each house and locality. Most of the waste generated is thrown into garbage bins, from where they are collected by the local municipalities [2]. This constant increase in solid waste is leading to global concern about future environmental problems [3] related to public health [4], agriculture, and education. As a result of the aforementioned problem, there is concern and dissatisfaction among the population with their municipal entities [5], which incur high costs in transportation (fuel, vehicle maintenance and spare parts) and operations (labor, uniforms and safety implements) in solid waste collection [6], which translates into an approximate 10% annual growth in operating costs [7]. It is one of the challenges that municipalities face.

Currently, several papers study solid waste management. Among them, we can mention the intelligent management system for real-time monitoring of ”garbage bins”, where they propose an intelligent bin mechanism based on IoT technology and applications and real-time monitoring of garbage bins [4]. Also, another IoT system that fully exploits cloud computing technology by developing a waste management system based on Deep Learning to revolutionize the current municipal waste management system [8]. Additionally, Machine Learning models and algorithms allows for accurate estimation of the amount of urban solid waste produced in different timeframes, for example, using the Seasonal ARIMA models (s-ARIMA) and Extreme Gradient Boosting (XGBoost) [9]. Likewise, some papers make use of the K-means algorithm to classify different zones into various groups and help analyze the characteristics of urban solid waste generation [5,10]. However, the aforementioned contributions do not combine the real-time IoT sensors and Machine Learning techniques, with the objective of observing daily generation of solid waste by zones.

Therefore, our proposal is to develop a tool consisting of predicting the amount of solid waste and generating the day’s collection plan, which will have the following layers: (1) IoT modules (capture layer), (2) system cluster that contains the data collection service (communications layer), forecast service (business logic layer) and the administrative module (presentation layer), (3) data cluster (storage layer). All layers and modules described run on the AWS cloud.

This work is organized as follows: In Section 2, a literature review or related work is carried out. Section 3 presents the proposed model. The validation of the study is carried out in

Section 4. The results and discussion are shown in Section 5. Finally, conclusions and future work are presented in Section 6.

II. Related Work

For the analysis of related works, a systematic literature review was conducted [10], where the following phases were applied: 1) Review planning, 2) Review development, and 3) results and analysis.

Starting with 4 questions related to components, technology, algorithm, and validation of solid waste collection plan prediction: What components do the systems for solid waste collection plan prediction have? (P1), What technologies can be used to build a system for solid waste collection plan prediction? (P2), What Machine Learning algorithms have been applied to solid waste collection plan prediction? (P3), How is a solid waste collection plan prediction system validated? (P4).

Subsequently, the search was conducted in scientific repositories such as Scopus, Web of Science, and Sciencedirect. Keywords used for this purpose included IoT sensors, predictive algorithms, path , Machine learning, and solid waste. The selection of articles depended on them being no more than 3 years old, belonging to quartile 1 or 2, and addressing predictive algorithms, route , IoT sensors, solid waste, and machine learning.

Likewise, for the analysis of the articles, a taxonomy related to each of the initially posed questions was used (Table I).

TABLE I
Articles According to Taxonomy.

Taxonomy	Source
Components (P1)	[3,4,9], [11-18]
Technologies (P2)	[1,4,11,13,16], [17-20]
Algorithms (P3)	[2,9,14,16,21,22]
Validation (P4)	[9,12,21]

As for the 'components', eleven were found, which are: Route [11], Machine learning model [9,11], Number of required vehicles [12], Well-being of drivers and customers (neighbors) [13], Cloud computing [4,8], Shared Socioeconomic Pathways [5], Microcontroller, ultrasonic sensor, real-time clock, battery, Citizen well-being [14], QSB software [8,15], Sustainable city [3,16], LoRa chip, and casing [17], Causal diagram and Forrester diagram for solid waste management [18]. The best component is the ultrasonic sensor because it can detect obstacles within a 15° angle and at a distance of 2 to 400 cm, and its results are stable and relatively accurate. The ultrasonic sensor is a component of our IoT device that reports the amount of solid waste to the solid waste collection plan prediction system.

On the other hand, regarding the 'technologies' found, four were obtained, which are: ArcGIS Network Analyst [17], CPLEX Optimizer [13], IoT (Internet of things) [4,11,16,19,20]. The most relevant for our study is IoT, which refers to the collective network of connected devices and the technology that facilitates communication between devices and the cloud. Along with Machine Learning, it will provide real-time data, enabling solid waste prediction and management for the municipality. Additionally, we used TensorFlow technology to conduct tests on our predictive model.

Similarly, concerning the 'algorithms' found, fourteen were obtained, which are: The K-means method [2,14], Hill Climber Algorithm [2], Recurrent Neural Network (RNN) [21], Long Short-Term Memory (LSTM) [21], SVM (Support Vector Machine), RF (Random Forest) [9], XGBoost [22], Path Cheapest Arc [14], Tabu Search algorithms, Greedy Descent [14], Mixed-Integer Linear Programming (MILP) [13], MOG-WOA (Multi-Objective Whale and Genetic Algorithms) [16], SARIMA [9], SDM Algorithm, time series forecast model, ARIMA (Autoregressive Integrated Moving Average) [22]. We conclude that the time series model with the ARIMA model and RF (Random Forest) are the best options for developing the solid waste forecast model because they work with datasets of few variables and deliver highly accurate results. Therefore, to develop the solid waste collection plan management system, we will develop and train a machine learning model using time series with the ARIMA model, which will help the system analyze and interpret historical solid waste data from 2003 to 2020.

Regarding 'validations', the most highlighted topics were method or tool comparisons [23], test scenarios [24], experimentation between AS-IS and TO-BE [25], minimizing solid waste management costs [15,21], and benchmarking between predictive models for the quantity of solid waste [9]. Therefore, to validate the proposed tool, we will use experimentation between AS-IS and TO-BE.

The ultrasonic sensor component measures the available height within a container [25], IoT technology integrates electronic components to the cloud [25], the ARIMA algorithm forecasts an amount based on historical data [9], and validation comparing AS-IS and TO-BE demonstrates the benefit of the proposed solution. Through the integration of components, technology, algorithms, and validation, the proposed tool for predicting the quantity of solid waste and optimizing collection plans is developed and implemented. Finally, we can indicate that the use of IoT with Machine Learning helps municipalities manage daily plans and solid waste in different areas.

III. Proposed Tool

In this section, we present at a high level how the tool for solid waste collection plan prediction and improvement of management was developed, applying time series and AWS. This tool will enable the forecasting of the quantity of solid waste and generate an optimal collection plan by zones with the objective of saving operational costs and reducing collection time, particularly in the public cleaning sector.

Below is a high-level diagram of the proposed solution for the municipality, utilizing the IoT system for the prediction of solid waste collection plans. The activities optimized by the tool are highlighted in blue (see Fig. 1).

This tool consists of three layers: (1) IoT modules (capture layer), (2) the system cluster containing the data collection ser-

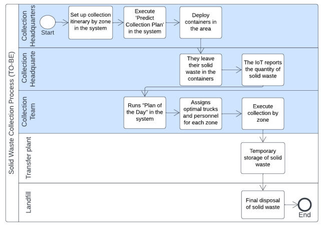

Fig. 1. A high-level TO-BE diagram

vice (communication layer), forecasting service (business logic layer), and the administrative module (presentation layer), and (3) the data cluster (storage layer). Figure 2 illustrates how these three layers collaborate in predicting solid waste collection plans. The IoT modules in the capture layer continuously monitor and report the quantity of solid waste in real-time to the system cluster, which receives, processes, converts the data into information for the solid waste predictive model, and presents it to end-users (solid waste collection managers) through the administrative module. Importantly, all solid waste data is stored in the data cluster for persistence.

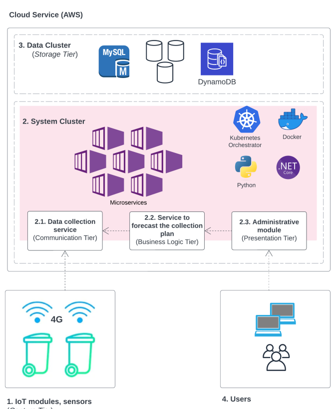

Fig. 2. Layer Integration in AWS

Below, we describe each layer of the system architecture:

A. IoT Modules, Sensors (Capture Layer)

This layer aims to capture data related to the quantity of solid waste through IoT devices installed within the containers. To achieve this, the following activities will be performed: (1) define IoT components, (2) configure and integrate the components, (3) install the integrated device inside the container, and (4) conduct testing.

Figure 3 shows the components that make up this layer, such as: A) Solar Panel, B) Lithium Battery, C) Voltage Booster, D) Ultrasonic Sensor HC-SR04, E) 4G Chip, F) Load Cell Sensor Type S YZC-516C.

A. Solar Panel: Provides power to the system's battery.

B. Lithium Battery: Powers the Arduino-Uno microcontroller.

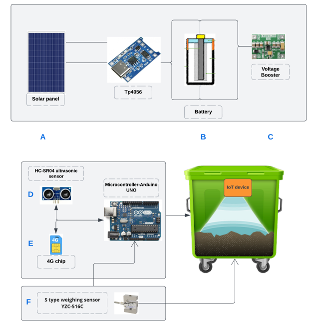

Fig. 3. Components of the IoT Sensor.

C. Voltage Booster: A low-profile, high-efficiency switching power source used to adapt the supply voltage at different stages of a circuit and power logic circuits from batteries.

D. Ultrasonic Sensor HC-SR04: Scans the quantity of solid waste inside the container. It can detect obstacles within a 15° angle and at a distance of 2 to 400 cm, providing stable and relatively accurate results.

E. 4G Chip: Through 4G technology, the IoT device accesses the internet to report the quantity of solid waste to the cloud.

F. Load Cell Sensor Type S YZC-516C: This weight sensor features a price-plant S-type load cell, a measuring range of 100 kg 2̃T, voltage and compression force options, good stability, and low temperature.

The main types of sensors are height (ultrasonic) and weight sensors. The first determines the height percentage (content within the container), and the second measures the weight of the content in kilograms. The Arduino microcontroller sends these data (via the 4G network) to the system cluster for processing and feeding the predictive model.

B. System Cluster

We built the server cluster that will run the system's services and modules through the following phases: (1) choosing the cloud, (2) deploying the services in the cloud, (3) configuring and testing the system's services.

We have chosen AWS cloud to run the system cluster because, unlike other providers (Azure, Google, IBM, and Oracle), it offers lower costs, better technical support, better integration with IoT devices (through its IoT Core service), and better management of the Kubernetes technology we are using as the cluster orchestrator.

The system's services and modules were developed using open-source software such as Python, MySQL, .NET Core, and Visual Studio Code. This approach allows the system to scale horizontally without cost restrictions related to licenses. The system's cluster includes the following services: 1. Data collection service (communication layer), 2. Service for predicting the collection plan (business logic layer), and 3. Administrative module (presentation or application layer).

Figure 4 illustrates how the entire system has been deployed. The system requires a client PC with an internet browser, which connects to the cloud (AWS) via the internet. In the cloud, it runs the MQTT service (communication layer), the system cluster (executed by Kubernetes), and the data cluster.

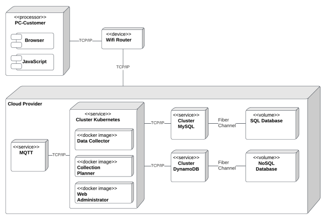

Fig. 4. Deployment View Diagram.

Data Collection Service (Communication Layer):

The data collected by IoT devices in the capture layer report the quantity of solid waste to the MQTT message queue in the cloud (see Fig. 5), and this queue reports new messages in real-time to the Data Processor service (queue subscriber). The Data Processor service records historical data and reports only data variations to the waste collection planner and web administrator services.

Fig. 5. Communication between the IoT Device and the Cloud

Service for forecasting the collection plan (Business Logic Layer): This service is responsible for predicting the quantity of solid waste that one or more areas may produce in the future. The forecast is based on time series with the ARIMA model in Python, which was trained with a data set of monthly quantities of solid waste from 2003 to 2020, as well as the data provided by the communication layer. The data from 2003 to 2017 was used to train the model, and the data from 2018 to 2020 was used to test the model and verify its level of confidence (see Fig. 6).

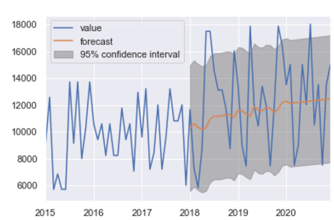

Fig. 6. Predictive Model Confidence Interval

The ARIMA model was chosen over RF (Random Forest) because it does not require many features (input variables). In this study, we only had access to the quantities of solid waste, not other socioeconomic aspects that can also influence the forecast of the quantity of solid waste. Additionally, IoT devices send the quantity of solid waste (stored in containers) to the model every hour. These data are used to constantly train the model for a maximum of one year, after which the model must be analyzed and validated again to confirm that the forecast still maintains an adequate level of accuracy (75% or more).

Administrative Module (Presentation or Application Layer): Responsible for providing application services to end-users, it integrates with the communication and business logic layers to offer three functionalities: (A) plan collection itinerary, (B) daily collection plan, and (C) predict collection plan. Figure 7 shows the system menu and the option for the daily plan.

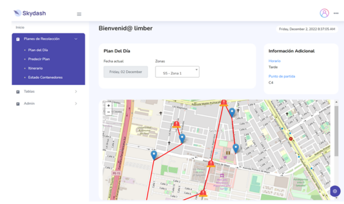

Fig. 7. Administrative Web

Additionally, the Collection Planner depends on the Data Collector to obtain historical data and train the predictive model. Finally, the Web Administrator relies on the Collection Planner to obtain the solid waste forecast and use it in collection planning functions for end users [16]. Below is a description of the process of the "Predict Collection Plan" functionality (presented later in the validation section).

Predict Collection Plan: Predicts the average amount of solid waste that an area (or multiple areas) will generate within a range of future dates. This is integrated with the "Collection Planner" service.

- The Predict Plan screen controller receives the request for one or more zones and a future date range. It forwards this query to the Business Logic.
- The Business Logic obtains the forecasted quantity of solid waste by zone(s) and containers, using the "Collection Planner" service.
- It retrieves the available itineraries (shifts and trucks) for the zone. If none exist, the process is aborted.
- It retrieves the containers in the zone and processes them (determining which ones should be collected based on the forecast obtained in step 2 and discarding those that do not apply).
- It returns to the Controller for each zone: the forecast (total and by container) and the number of trucks required for the zone.

C. *Data Cluster (Storage Layer)*

Responsible for data persistence throughout the system, it stores transactional and analytical data generated and consumed by various components and services of the system. This layer is deployed within AWS RDS and NoSQL services, where, based on the following benchmarking, we chose to use MySQL as the transactional relational database and DynamoDB as the analytical NoSQL database, primarily due to cost-effectiveness and scalability (Table II).

TABLE II
BENCHMARKING OF AWS DATA CLUSTER COSTS.

Service	Initial cost	Monthly cost
Amazon RDS para MySQL	0.00 USD	196.70 USD
Amazon RDS para PostgreSQL	0.00 USD	204.00 USD
Amazon DocumentDB	0.00 USD	56.94 USD
Amazon DynamoDB	180.00 USD	26.64 USD

The data cluster utilizes the following database technologies:

- DynamoDB (NoSQL): Stores analytical data (quantity of solid waste by containers) constantly received from IoT devices
- MySQL (Relational Database): Stores information about zones, itineraries, containers, and trucks. This information is consumed by the application, as shown in Figure 4 (Deployment View Diagram).

IV. VALIDATION

The validation of this tool was carried out at the District Municipality of Villa El Salvador, a district in the south of Lima. Specifically, it was conducted in the Sub-Management of Public Cleaning and Cleaning for the Solid Waste Collection service.

To validate the proposed IoT tool, two experiments were conducted. In the first experiment, we observed the manual process of solid waste collection plan management at the Municipality of Villa El Salvador. In the second experiment, we tested the IoT tool (automation of collection plans) in Sector 5 of Villa El Salvador. In both experiments, we evaluated costs, collection time, and the experience of end users (Table III).

TABLE III
EXPERIMENTS

Experiment	Features to Evaluate	Scenery
· Experiment 1 (AS-IS) · Experiment 2 2 (TO-BE)	· Costs · Collection time · Neighbor satisfaction	· Sector 5 of Villa El Salvador

A. Experiment 1: Observation of the manual process

The objective is to determine the optimal number of garbage compactors to a specific sector, in this case, the experiment was conducted in sector 5 of Villa El Salvador, a residential area where garbage collection occurs during the night (see Fig. 8).

The Submanagement of Public Cleaning and Maintenance reports that Sector 5 generates approximately 9% of the district's solid waste. In its collection plan, it has allocated 3 compacting trucks, each with a loading capacity of 10 tons, providing a total collection capacity of 30 tons for Sector 5.

Fig. 8. Map of Villa el Salvador Sectors – Sector 5

Fig. 9. Compacting Trucks

We observed the use of compacting trucks for 2 weeks in sector 5. Additionally, we conducted a survey with a group of residents from that sector to measure their satisfaction with the solid waste collection process during these two weeks.

B. Experiment 2: Using the plan prediction tool

The validation of the proposed tool was carried out by implementing an IoT prototype and the management software deployed in the AWS cloud.

Hardware Configuration: The IoT device was assembled with a strain gauge weight sensor, HC-SR04 ultrasonic level (height) sensor, and the Arduino ESP32 microcontroller. The microcontroller has a built-in WIFI module that can transmit the captured weight and height information of solid waste to the AWS cloud. Figures 10 and 11 show the two sensors used in the experiment to capture the height and weight of solid waste inside the container.

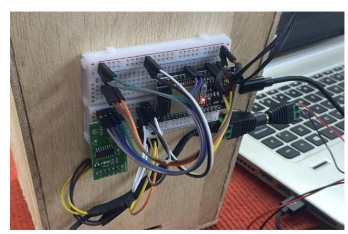

Fig. 10. ESP32 Microcontroller and Height Sensor (Ultrasonic).

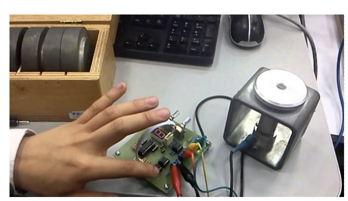

Fig. 11. Weight sensor.

Software Configuration: To configure the management software, we loaded the monthly dataset of solid waste from

2003 to 2020 (see Fig. 12), which is essential input for the predictive model (ARIMA algorithm) of the software. Additionally, we recorded the compacting trucks, the sector, their containers, and the itinerary in the system's database. The

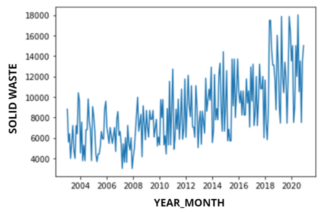

Fig. 12. Villa El Salvador solid waste data set

IoT prototype continuously sends the weight (amount of solid waste) and the height of solid waste inside the container (see Fig. 13). These data feed into the predictive model (retraining) and the administrative module, the functionality of the Daily Plan.

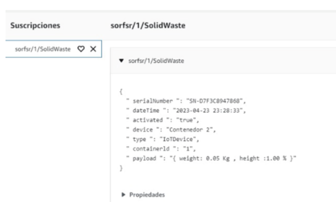

Fig. 13. AWS IoT Core activity log

The experiment primarily focused on using the Predict Plan functionality of the administrative module, for the same date range as Experiment 1 (see Fig 14).

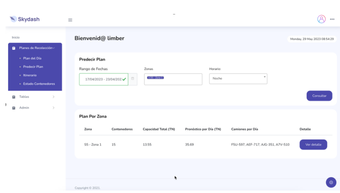

Fig. 14. Predict Plan Screen

V. Results and Discussion

The tool (Predict Plan functionality) indicates that for both weeks, the forecast for solid waste in the area is 35.69 tons per day, and the plan involves using 4 compacting trucks for the area during that period (each truck can carry a maximum of 10 tons).

To analyze and interpret these results, Table IV shows the comparison between manual estimation, actual collection, and the system forecast for the second week of the experiment.

Where the Precision of the estimation is calculated through a simple rule of three, comparing the estimation (manual or from the system) with what is actually collected. This was defined jointly with the municipality so that they can regularly validate the tool, avoiding complexity.

TABLE IV
Manual Estimation vs. System Forecast

Day	Manual Estimation	Real Collected	System Forecast
Monday	30	37	35.69
Mars	30	30	35.69
Wednesday	30	37	35.69
Thursday	30	31	35.69
Friday	30	33	35.69
Saturday	30	32	35.69
Sunday	30	27	35.69
Total Tons	**210**	**227**	**249.83**
Average	**30**	**37 (maximum)**	**35.69**
Precision	**0.81**		**0.96**

The results for week 2 are summarized in the following bar chart, where we can compare the accuracy of the system's forecast vs. manual estimation in tons. Manual estimation has an accuracy of 81% (30/37), and the system's solid waste quantity forecast is 96% (35.69/37). This shows that we have achieved better efficiency in waste collection using optimal routes, which saves fuel and time on the trucks (see Fig. 15).

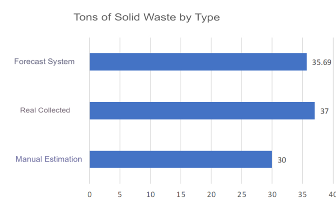

Fig. 15. Solid Waste Comparison

The system reports the optimal number of waste collection trucks to send to an area. This directly influences the operating costs of solid waste collection (Table V). Where the manually estimated plan is unsuccessful (it does not collect the actual amount of solid waste, 37 tons) and generates an additional cost for rework (sending an extra truck and personnel, overtime hours).

TABLE V
Impact on Operating Costs

Comparative	Manual Estimation	System Forecast
Average tons	30	35.69
Suggested trucks (plan)	3	4
Daily cost per truck (fuel and maintenance)	S/ 120	S/ 120
Daily cost of personnel (driver and two operators)	S/ 200	S/ 200
Daily cost	S/ 960	S/ 1,280
Plan is successful	No	Yes
I cost rework	S/ 432	S/ 0
Actual daily cost	S/ 1,392	S/ 1,280

Through the use of the system, we not only optimize costs but also save time by avoiding rework in solid waste collection. The "Plan del Día" functionality allows us to map the optimal collection route (using real-time data from IoT containers), avoiding streets where there is not much solid waste generated.

Achieving optimal solid waste collection in terms of costs and time results in high satisfaction among operators and residents.

Efficient routes have a positive and efficient impact on institutions through the combination of IoT and algorithm training. However, it can also become a limitation if one of the components fails.

VI. Conclusions And Future Work

Through IoT technology, AWS cloud, a predictive algorithm, and custom software development, we have achieved a tool for predicting the best solid waste collection plan, allowing us to save time and resources and achieve better efficiency in the municipality of Lima.

During the validation of the tool, we have achieved up to 96% accuracy in predicting the amount of solid waste that a district area can generate. Based on this, we can calculate the number of compacting trucks to deploy in one or more district areas to achieve optimal solid waste collection, saving operational costs and reducing collection time.

IoT technology helped us to continuously monitor the amounts of solid waste stored in the containers at short intervals. These data, constantly received in the AWS cloud, allowed us to determine the best collection route and set the foundation for future comparisons between the predicted collection plan and the actual amounts of solid waste reported by the IoT containers.

In the field of machine learning, we used the ARIMA statistical model and a 17-year solid waste dataset from the Villa El Salvador district. We developed a predictive algorithm in Python that analyzes variations and regressions in statistical data to find patterns in monthly solid waste generation, thus predicting future solid waste quantities in the district.

As future work, we recommend testing the tool in other municipalities in the Lima Metropolitan area in various district zones with their respective datasets. One of the significant challenges is obtaining information from IoT devices that will serve as input for the algorithm training. Additionally, extending the tool with reports to compare the effectiveness of plan predictions with real-world information is important. Furthermore, evaluating more sophisticated predictive models like Gradient Boosting, which requires additional datasets such as population, economically active individuals, industries, hospitals, commercial movement in markets, and other socioeconomic data by month over several years.

Acknowledgment

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